

Technical Documentation

Diesel Engine
16 V 595 TE90

Operating Instructions
M015286/00E



Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

Diese Veröffentlichung einschließlich aller ihrer Teile ist urheberrechtlich geschützt. Jede Verwertung oder Nutzung bedarf der vorherigen schriftlichen Zustimmung der MTU Friedrichshafen GmbH. Das gilt insbesondere für Vervielfältigung, Verbreitung, Bearbeitung, Übersetzung, Mikroverfilmungen und die Einspeicherung und / oder Verarbeitung in elektronischen Systemen, einschließlich Datenbanken und Online-Diensten. Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen. Änderungen bleiben vorbehalten.

Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Imprimé en Allemagne

© 2006 Copyright MTU Friedrichshafen GmbH

Tout droit réservé pour cet ouvrage dans son intégralité. Toute utilisation ou exploitation requiert au préalable l'accord écrit de MTU Friedrichshafen GmbH. Ceci s'applique notamment à la reproduction, la diffusion, la modification, la traduction, l'archivage sur microfiches, la mémorisation et / ou le traitement sur des systèmes électroniques, y compris les bases de données et les services en ligne.

Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

© 2006 Copyright MTU Friedrichshafen GmbH

Esta publicación se encuentra protegida, en toda su extensión, por los derechos de autor. Cualquier utilización de la misma, así como su reproducción, difusión, transformación, traducción, microfilmación, grabación y/o procesamiento en sistemas electrónicos, entre los que se incluyen bancos de datos y servicios en línea, precisa de la autorización previa de MTU Friedrichshafen GmbH.

El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

Nos reservamos el derecho de introducir modificaciones.

Stampato in Germania

© 2006 Copyright MTU Friedrichshafen GmbH

Questa pubblicazione è protetta dal diritto d'autore in tutte le sue parti. Ciascun impiego o utilizzo, con particolare riguardo alla riproduzione, alla diffusione, alla modifica, alla traduzione, all'archiviazione in microfilm e alla memorizzazione o all'elaborazione in sistemi elettronici, comprese banche dati e servizi on line, deve essere espressamente autorizzato per iscritto dalla MTU Friedrichshafen GmbH.

Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

Impresso na Alemanha

© 2006 Copyright MTU Friedrichshafen GmbH

A presente publicação, inclusive todas as suas partes, está protegida pelo direito autoral. Qualquer aproveitamento ou uso exige a autorização prévia e por escrito da MTU Friedrichshafen GmbH. Isto diz respeito em particular à reprodução, divulgação, tratamento, tradução, microfilmagem, e a memorização e/ou processamento em sistemas eletrônicos, inclusive bancos de dados e serviços on-line.

Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

Wichtig – Important – Importante

Bitte die Karte „Inbetriebnahmemeldung“ abtrennen und ausgefüllt an MTU Friedrichshafen GmbH zurücksenden.

Die Informationen der Inbetriebnahmemeldung sind Grundlage für den vertraglich vereinbarten Logistik-Support (Gewährleistung, Ersatzteile etc.).

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).

Veillez séparer la carte “Signalisation de mise en service” et la renvoyer à la MTU Friedrichshafen GmbH.

Les informations contenues dans la signalisation de mise en service constituent la base pour l'assistance en exploitation contractuelle (garantie, rechanges, etc.).

Rogamos separar la tarjeta “Aviso de puesta en servicio” y la devuelvan rellena a MTU Friedrichshafen GmbH.

Las informaciones respecto al aviso de puesta en servicio constituyen la base para el soporte logístico contractual (garantía, piezas de repuesto, etc.).

Ritagliare “Avviso di messa in servizio” e rispedirlo debitamente compilato alla MTU Friedrichshafen GmbH.

Le informazioni ivi registrate sono la base per il supporto logistico contrattuale (garanzia, ricambi, ecc.).

É gentileza cortar o cartão “Participação da colocação em serviço”, preenché-lo e devolvê-lo a MTU Friedrichshafen.

Os dados referentes à colocação em serviço representam a base para o suporte logístico (garantia, peças sobressalentes, etc.) estabelecido contratualmente.

	Postcard MTU Friedrichshafen GmbH Department SCSD 88040 Friedrichshafen GERMANY
---	--

Bitte in Blockschrift ausfüllen!
Please use block capitals!
Prière de remplir en lettres capitales!
¡A rellenar en letras de imprenta!
Scrivere in stampatello!
Favor preencher com letras de forma!



Motomr.: Engine No.: N° du moteur: N° de motor: Motore N.: No. do motor:

Auftragsnr.: MTU works order No.: N° de commande: N° de pedido: N. commessa: No. do pedido:
--

**Inbetriebnahme-
meldung**

**Commissioning
Note**

**Notice de mise
en service**

**Aviso de puesta
en servicio**

**Avviso di messa
in servizio**

**Participação da
colocação em
serviço**

Motortyp: Engine model: Type du moteur: Tipo de motor: Motore tipo: Tipo do motor:

Inbetriebnahmedatum: Date put into operation: Mise en service le: Fecha de puesta en servicio: Messa in servizio il: Data da colocação em serviço:

Eingebaut in: Installation site: Lieu de montage: Lugar de montaje: Installato: Incorporado em:
--

Schiffstyp / Schiffshersteller: Vessel/type/class / Shipyard: Type du bateau / Constructeur: Tipo de buque / Constructor: Tipo di barca / Costruttore Tipo de embarcação/estaleiro naval:
--

Endabnehmer/Anschrift: End user's address: Adresse du client final: Dirección del cliente final: Indirizzo del cliente finale: Usuário final/endereço:

Bemerkung: Remarks: Remarques: Observaciones: Commento: Observações:

1	Safety	05
1.1	General conditions	05
1.2	Personnel and organizational requirements	06
1.3	Transport	07
1.4	Safety precautions when working on the engine	08
1.5	Auxiliary materials, fire prevention and environmental protection	11
1.6	Standards for warning notices in the publication	13
2	Product Summary	15
2.1	Engine Layout	15
2.1.1	Engine layout	15
2.2	Engine Side and Cylinder Designations	17
2.2.1	Engine side and cylinder designations	17
2.3	Main Engine Dimensions	18
2.3.1	Main engine dimensions	18
2.4	Firing Order	19
2.4.1	Firing order	19
2.5	Technical Data	20
2.5.1	12V 595TE90, 16 V595TE90 engine data	20
3	Operation	23
3.1	LOP – Overview	23
3.2	Pump control – Control and display elements	30
3.3	Putting the engine into operation after extended out-of-service periods (>3 months)	31
3.4	Putting the engine into operation after scheduled out-of-service periods	33
3.5	Starting the engine from LOP	34
3.6	Operational checks	36
3.7	Stopping the engine from LOP	37
3.8	Emergency stop	38
3.9	After stopping the engine	40
3.10	PPC, LPU – Functional check	41
4	Maintenance	43
4.1	Preface	43
4.2	50-hours service	45
4.3	Maintenance schedule matrix	46
4.4	Maintenance tasks	50
5	Engine Control System ECS-5	53

5.1	Normal Operation: LOS or LOP with Display	53
5.1.1	Local control mode – Engaging/disengaging the gearbox	53
5.1.2	Local control mode – Changing engine speed	54
6	Troubleshooting	55
6.1	Troubleshooting	55
6.2	Fault indication and fault messages	64
6.3	Oil priming pump, oil refill pump – Troubleshooting	92
7	Task Description	95
7.1	Engine	95
7.1.1	Engine – Barring manually	95
7.1.2	Engine – Barring with starting system	97
7.2	Cylinder Liner	98
7.2.1	Combustion chambers – Draining condensate	98
7.3	Valve Gear	99
7.3.1	Valve clearance – Check and adjustment	99
7.3.2	Cylinder-head cover – Removal and installation	101
7.4	Injector	102
7.4.1	Injector – Cleaning	102
7.4.2	Injector – Check	103
7.4.3	Injector – Replacement	105
7.4.4	Injector – Removal and installation	106
7.5	Fuel System	112
7.5.1	Leak-fuel tank – Draining	112
7.5.2	Fuel system – Venting	113
7.6	Fuel Filter	114
7.6.1	Fuel filter – Replacement	114
7.6.2	Fuel prefilter – Differential pressure check and adjustment of gauge	116
7.6.3	Fuel prefilter – draining	117
7.6.4	Fuel prefilter – flushing	119
7.6.5	Fuel prefilter – Filter element replacement	121
7.7	Charge-air Cooling, General, Left Side	124
7.7.1	Intercooler – Checking condensate drain line for coolant discharge and obstruction	124
7.8	Air Filter	125
7.8.1	Air filter element – Cleaning	125
7.8.2	Air filter replacement	126
7.8.3	Air filter removal and installation	127
7.9	Air Intake	128
7.9.1	Air intake flap – Lubrication	128
7.9.2	Service indicator – Signal ring position check	129
7.9.3	Charge-air manifold – Checking for coolant leakage	130
7.9.4	Emergency air shut-off flap – Lubrication	131
7.9.5	Emergency air shut-off flaps – Electric control check	132
7.9.6	Emergency air shut-off flaps – Opening/closing manually	133

7.10	Exhaust Flap with Actuators	134
7.10.1	Flap control – Functional test	134
7.11	Starting Equipment	135
7.11.1	Pressure reducing station – Putting into operation	135
7.11.2	Pressure reducing station – Venting and putting out of operation	137
7.11.3	Pressure reducing station – Filter cleaning	138
7.11.4	Starter – Condition check	139
7.12	Lube Oil System, Lube Oil Circuit	140
7.12.1	Engine oil – Change	140
7.12.2	Engine oil – Level check	142
7.12.3	Engine oil – Sample extraction and analysis	144
7.12.4	Oil sludge inspection for metallic residues	145
7.12.5	Lube oil priming pump – Checking direction of rotation	146
7.13	Oil Cooling / Filtration	147
7.13.1	Disk-pack filter – Cleaning	147
7.13.2	Engine oil filter – Draining oil sludge	148
7.13.3	Engine oil filter – Cleaning filter element	149
7.13.4	Engine oil filter – Replacement	150
7.13.5	Centrifugal oil filter – Cleaning and filter-sleeve replacement	151
7.13.6	Engine oil bypass filter – Replacement	154
7.14	Coolant Circuit General, HT Circuit	156
7.14.1	Drain, venting and measuring points	156
7.14.2	Vent line – Checking for obstructions	161
7.14.3	Engine coolant – Change	162
7.14.4	Engine coolant system – Filling	163
7.14.5	Engine coolant – Draining	164
7.14.6	Engine coolant – Level check	165
7.14.7	Engine coolant pump – Relief bore check	166
7.14.8	Engine coolant – Sample extraction and analysis	167
7.14.9	HT circuit coolant thermostat – Emergency adjustment	168
7.14.10	HT coolant circuit – Strainer removal	169
7.15	LT Circuit	170
7.15.1	LT circuit coolant thermostat – Emergency adjustment	170
7.16	Raw Water Pump with Connections	171
7.16.1	Raw water pump – Relief bore check	171
7.17	Engine Mounting	172
7.17.1	Engine mounts – Check and adjustment	172
7.17.2	Engine mounts – securing screws check	175
7.18	Wiring, General for Engine / Gearbox / Plant	176
7.18.1	Engine wiring check	176
7.18.2	Sensors and actuators	177
7.19	Accessories for (Electronic) Engine Governor / Control System	183
7.19.1	ECS – Connectors and housing cleaning	183
7.19.2	Plug-in connections – Check	184
7.19.3	ECS – Removal and installation	185

7.20	General Display and Control Elements	186
7.20.1	Limit switch – Check	186
7.20.2	Indicator lamp - LED replacement	187
7.20.3	Fluid level monitor – Check	189
7.21	Emergency Instrumentation (Control Stand)	190
7.21.1	Engine control system – Operational tests	190
8	Special Tools	193
8.1	Special tools	193
8.2	Standard tools – Overview	200
9	Annex	201
9.1	Manufacturer's documentation	201
9.2	MTU contact person/service partner	202
9.3	Index	203

1 Safety

1.1 General conditions

General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed. This engine is a state-of-the art product and conforms with all applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Non-compliance with the Safety Instructions

Correct use

The engine is intended exclusively for the application specified in the contract or defined at the time of delivery. Any other use is considered improper use. The manufacturer will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the maintenance specifications.

Modifications or Conversions

Modifications made by the customer to the engine may affect safety.

MTU will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

Spare parts

Only genuine MTU spare parts must be used to replace components or assemblies. In the event of any damage caused by the use of other spare parts, no liability nor warranty claims vis-à-vis the engine manufacturer will be accepted.

1.2 Personnel and organizational requirements

Personnel

Work on the engine must only be carried out by properly qualified and instructed personnel.

The specified legal minimum age must be observed.

Responsibilities of the operating, maintenance and repair personnel must be specified.

Organization

This publication must be issued to all personnel involved in operation, maintenance, repair or transportation.

It must be kept at hand near the engine and accessible at any time to all personnel involved in operation, maintenance, repair or transportation.

The personnel must be instructed on engine operation and repair by means of this publication, and in particular the safety instructions must be explained.

This is especially important for personnel who work on the engine only on an occasional basis. Such personnel must be given instructions repeatedly.

Working clothes and protective equipment

Wear proper work clothing for all work.

Depending on the kind of work, use additional protective equipment, e.g. protective goggles, gloves, helmet, apron.

Work clothing must be tight fitting so that it does not catch on rotating or projecting components.

Do not wear jewelry (e.g. rings, chains etc.).